Seasonal Weighing and Moisture Monitoring of the Heritage Wooden Fleet: A Field Method

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ABSTRACT. Aston Ferry Rowing Club keeps nine wooden boats built between 1958 and 1994 for exhibition races and heritage events. Water uptake in the hull timber is the earliest measurable sign of failing varnish and of the soft rot that ended the life of the club's 1952 eight in 2019. This note sets out how the fleet is weighed with two suspended load cells, how results are compared with each boat's reference dry mass, how pin moisture readings are corrected for timber species, and who acts on the Amber and Red levels.

1. Purpose and scope

The method applies to the nine boats in Table 1, stored in Bay 3 of the boathouse on the Berkshire bank opposite Temple Island. It does not apply to the club's 31 composite boats or to the launches, which have their own survey (AFRC-MN-03).

Each boat is weighed four times a year: in the first fortnight of March, in the week after Henley Royal Regatta, in mid-September and in the first week of December. A boat lent to an outside event (for example a film shoot at Hambleden) is also weighed within 10 days of its return. A reading taken outside the conditions in Section 3 is recorded as “non-standard” and never used for an Amber or Red decision.

2. Equipment and calibration

Version 1.4 of this note used two platform scales under the canvas, which gave a spread of up to 1.8 kg on repeat weighings of the same eight. From the September 2026 weighing onward the boat is lifted clear of the trestles in two 75 mm webbing slings, each hanging from a Kestrelscale KS-300 crane load cell (0–300 kg, resolution 0.05 kg; club assets BC-114 bow and BC-115 stern) on the Bay 3 gantry. The total mass is the sum of the two readings minus the tare of the slings and shackles, which was measured on 2 September 2026 as 1.35 kg for the pair and must be remeasured whenever a sling or shackle is replaced.

Before every session each cell is zeroed with its sling hanging and the 20 kg cast-iron check mass (BC-117) is hung on it. If either cell reads outside 19.90–20.10 kg the session is postponed and the Boatman is told the same day; members never adjust the cells.

3. Conditions for a standard weighing

A hull weighed straight after an outing reads heavy because of surface water, so a weighing counts as standard only if all of the following apply:

- the boat has not been on the water for at least **72 hours** and has been stored upside down on its rack in Bay 3 for that whole period (version 1.4 required 48 hours);
- the bilge has been sponged out and the boat has been tipped on its side for 5 minutes to drain the canvas and buoyancy spaces;
- relative humidity in Bay 3 is **75 % or less** and air temperature is between **8 °C and 24 °C**, read from the Bay 3 thermo-hygrometer at the start;
- seats, stretchers, riggers and the cox box are removed and the boat is weighed bare; fixed fittings such as the rudder, fin and saxboard brass stay on.

If Bay 3 is too humid, single and double sculls only may be weighed in the gym loft, hanging from the loft beam with the same cells.

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4. Reference dry mass and action levels

Each boat's reference dry mass is the mean of the standard weighings of March and September 2023, after the fleet had been dried for six weeks by Rushbrook Wooden Boat Conservation at Wargrave. For carvel and moulded hulls the Amber level is a gain of 2.5 % and the Red level a gain of 5.0 %; for the single sculls, whose small mass makes the result more sensitive to fittings, the levels are 3.0 % and 6.0 %. Clinker hulls take up water through their laps by design, so for the two tubs the levels are 4.0 % and 7.0 %. In version 1.4 every boat used 3.0 % and 6.0 %. Table 1 gives the resulting masses in kilograms, rounded to 0.1 kg; the table values are the ones to apply.

Table 1. Heritage fleet: reference dry mass and action levels (version 2.1). Tub 2 = clinker tub pair, Tub 4 = clinker tub four

Boat	Class	Built	Hull	Ref. kg	Amber kg	Red kg
Marjorie Fane	8+	1958	Red cedar, carvel	118.6	121.6	124.5
Gwendolen	8+	1971	Spanish cedar, carvel	112.4	115.2	118.0
Sir Roger Lomax	4+	1964	Red cedar, carvel	61.8	63.3	64.9
Kingfisher II	4-	1979	Spanish cedar, carvel	53.2	54.5	55.9
Bellweather	2x	1983	Mahogany, moulded	31.5	32.3	33.1
Fennimore	1x	1987	Spanish cedar, moulded	15.9	16.4	16.9
Ada Treherne	1x	1994	Red cedar, moulded	14.7	15.1	15.6
Old Tom	Tub 2	1962	Larch on oak, clinker	96.0	99.8	102.7
Heron	Tub 4	1966	Larch on oak, clinker	148.5	154.4	158.9

4.1 What happens at each level

A result below the Amber level needs no action beyond recording. At or above the Amber level the boat stays in use, but the Boat Care Committee chair must be told within 7 days, the varnish is inspected for cracks, and the boat is weighed again after a further 14 days dry in Bay 3. Two Amber results in a row are treated as Red.

At or above the Red level the boat is withdrawn from all rowing at once: the weighing member hangs a red tag on the bow ball and locks the boat on its rack, and the Captain of Boats is informed the same day. Only the Captain of Boats may release the boat again, and only after Rushbrook Wooden Boat Conservation has surveyed the hull and a standard weighing below the Amber level has been recorded. A boat lent to an outside event while carrying a Red tag is not covered by the club's insurance.

A result is never corrected for new fittings. If a fitting has been added or replaced, the Boat Care Committee asks the Club Committee to revise the boat's reference in Table 1; until the revision is approved the old reference applies.

5. Procedure

Two members who have completed the half-day weighing course carry out each session; one must sit on the Boat Care Committee.

1. Record the date, RH, temperature and hours since the last outing, and do the check-mass test (Section 2).
2. Set the slings at the positions in Table 2, measured from the bow ball along the gunwale, with the spreader board inside the hull at each sling.
3. Lift the boat with the gantry chain hoists until it is 50 mm clear of the trestles, level it to within 20 mm end to end, and wait 60 seconds for the readings to settle.
4. Note both readings, lower, and lift twice more. The result is the mean of the three totals; if they differ by more than 0.30 kg, find the cause and repeat.
5. Take pin moisture readings at the six stations in Section 6.

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Table 2. Sling positions and gantry settings by boat class

Boat class	Bow sling (m)	Stern sling (m)	Spreader board	Hoist operators
Eight (8+)	4.60	12.90	Long, 1.2 m	2 + 4 carriers
Coxed four (4+)	3.10	9.20	Long, 1.2 m	2 + 2 carriers
Coxless four (4-)	2.90	8.70	Long, 1.2 m	2 + 2 carriers
Double scull (2x)	2.30	7.10	Short, 0.8 m	2
Single scull (1x)	2.00	6.10	Short, 0.8 m	2
Clinker tub pair	1.70	5.20	None (on keelson)	2
Clinker tub four	2.40	7.40	None (on keelson)	2 + 2 carriers

6. Moisture readings and species correction

Pin moisture readings are taken at six fixed stations: the keel at bow and stern canvas, and both garboards at the bow seat and at the stroke seat. The pins go 6 mm deep through brass witness plugs.

The meter is calibrated for Douglas fir, so its displayed reading is converted to a corrected moisture content with Table 3, prepared by Rushbrook from oven-dried offcuts. For readings between two columns, use the nearer column; for a reading exactly halfway, use the higher one. No temperature correction is applied between 8 °C and 24 °C.

Table 3. Corrected moisture content (%) for the meter's displayed reading, by hull timber

Hull timber	Shows 10 %	Shows 14 %	Shows 18 %	Shows 22 %	Shows 26 %
Western red cedar	11.2	15.6	20.1	24.5	28.8
Spanish cedar	10.6	14.9	19.0	23.2	27.3
Mahogany veneer	9.4	13.1	16.9	20.6	24.4
Larch planking	9.8	13.7	17.5	21.4	25.2
Oak keel and frames	9.1	12.6	16.2	19.8	23.5

A corrected moisture content below 16 % at all six stations is normal. A corrected value of 16 % to 19.9 % at any station is reported with the weighing and the station is rechecked at the next session. A corrected value of 20 % or more at any station is treated as a Red result for that boat even if its mass is below the Red level.

7. Recording, review and responsibilities

The Boatman, currently Dennis Okafor-Hale, keeps the equipment, books Bay 3 and the gantry for each session, and arranges external verification of the check mass. Owen Catchpole, as Boat Care Committee chair, reviews every Amber result and signs off the quarterly summary for the Club Committee. Harriet Ashdown-Vere, as Captain of Boats, decides on withdrawals and releases and is the only person who may approve the loan of a heritage boat for an outside event.

References

1. Aston Ferry Rowing Club (2022). *Weighing the heritage boats*, method note AFRC-MN-07, version 1.4.
2. Rushbrook Wooden Boat Conservation (2023). *Drying and baseline survey of the Aston Ferry heritage fleet*, report RWB-23/118. Wargrave.

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